CONTRIBUTION:	ESCO0836
DATE:	31 March 2026
WORKING GROUP:	Emergency Services (9-1-1) Working Group (ESWG)
TITLE:	Utilizing Alternative Addressing Systems to Standardize and Disambiguate Civic Locations for NG9-1-1
ISSUES ADDRESSED:	This contribution was created in response to ESCO0737, ESCO0748, and ESCO0785. It provides additional information and a methodology for representation of non-traditional civic locations in NG9-1-1.
SOURCE:	**Cynthia Mamer**, Saskatchewan Public Safety Agency – Emergency Communications **Natasha Roenspies**, Saskatchewan Public Safety Agency – Emergency Communications **Tanya Vokey**, Government of Newfoundland and Labrador, Department of Justice and Public Safety, Emergency Services – 911 Division **Robert Harris**, Co-chair ESTF0092
DISTRIBUTION:	CISC ESWG; ESTF0092
RELATED TIFS:	ESTF0085; ESTF0094; ESTF0095
FILE NAME:	ESCO0836_TIF92_Various - Alternate Address Standardization.pdf
NOTICE:	*This contribution has been prepared as basis for discussion in order to assist the Emergency Services Working Group. This contribution should not be construed as a binding proposal on the authors. Specifically, the authors reserve the right to amend, modify or withdraw this contribution at any time.*

Utilizing Alternative Addressing Systems to Standardize and Disambiguate Civic Locations for NG9-1-1.

Background

As local governments and authorities across Canada prepare for their transition to NG9-1-1 geospatial call routing, they are assessing the level of preparedness and completeness of their GIS data. NG9-1-1-compliant GIS data is the fuel that enables geospatial call routing. The requirements around structure and completeness of this data have been established by TIF92 but local governments must interpret these standards and adopt best practices to meet local geographies and settlement histories. Over the years, due to a variety of factors, many non-standard and non-authoritative civic locations have been created for civic addressing systems and emergency response/9-1-1 datasets. Whether these civic locations are known as non-traditional addresses, pseudo-addresses, or alternative addresses, they share a similar goal of needing a standardized methodology for inclusion in NG9-1-1-compliant GIS data, so that they can be used for NG9-1-1 services such as emergency call routing and location validation.

Goals

The NENA NG9-1-1 GIS Data Model and NENA NG9-1-1 CLDXF-CA standards aim to provide a way for all Canadian jurisdictions to describe civic locations in a standardized and unambiguous manner so that they can be leveraged in NG9-1-1 services, regardless of if they are currently in an E9-1-1, B9-1-1, or unserved area. There is a desire for GIS data providers to have a consistent methodology to ensure that they are representing their civic locations correctly. Additionally, it is understood that the effort to create a "complete" dataset of civic locations may never be achievable, but effort should be spent to ensure that, at a minimum, E9-1-1 areas have matching locations for existing valid records in the E9-1-1 ALI database.

The specific goals of this contribution are:

- Establish a standardized methodology to collect non-traditional locations
- Ensure methodology aligns with established NG9-1-1 GIS data requirements
- Assign an appropriate identifier to civic locations for emergency call routing
- Leverage NG9-1-1 GIS Data Model and CLDXF-CA as much as possible

    - CLDXF-CA defines the format and business rules for exchanging civic locations used in NG9-1-1
    - GIS Data Model defines the structure of GIS layers used in NG9-1-1
- Ensure an appropriate level of completeness of GIS data for emergency call routing
    - Wireless calls require service boundary polygons
    - Wireline calls require an identifier/geography and service boundary polygons
        - That geography can be either a SSAP feature or RCL address range
- Ensure civic locations can be disambiguated and validated consistently and cover all appropriate legacy ALI records.

Proposed Addressing Authority & GIS Data Provider Methodology:

This contribution proposed a 2-step process outlined in ESCO0785 and leverages the Site/Structure Address Point (SSAP) and Administrative Level (A1-A5) GIS data layers from the NENA NG9-1-1 GIS Data Model:

Step 1 - Disambiguate a place name

> Establish and complete as many Administrative Level elements as needed to disambiguate a place name to the most granular level possible, even if it is to a level more granular than applicable for your province or territory.

Step 2 - Disambiguate a location

> Ideally, all civic addresses would include either a Street Name and an Address Number, or a Street Name and a Distance Marker. Where this is not possible, effort should be made to satisfy one, or more, of the following conditions:
>
> - A Site whose identifier is unique within the given set of place names; or
> - A Structure whose identifier is unique within the given set of place names; or
> - A Location Marker whose identifier is unique within the given set of place names.

In addition to the GIS data methodology, a workflow will need to be established between the Originating Network Provider (ONP), the 9-1-1 authority/LGA, and the GIS data providers and addressing authorities that support them, to ensure that a 1:1 match remains between all valid endpoint wireline customer records and civic locations represented in NG9-1-1-compliant GIS data.

Saskatchewan History and Example usage:

The Saskatchewan Public Safety Agency (SPSA) manages a provincial Civic Address Registry (CAR) as per The Emergency 911 System Act - Section 5(c). This geospatial registry

database was created between 2010-2015 through the aggregation of authoritative addressing data from local governing authorities (LGA's), as well as the introduction of a unique rural addressing system that was made available for rural municipalities to adopt. The purpose of this registry was not only to provide better location data to the Provincial Emergency Communication Centre (PECC), a primary PSAP, but also to align GIS addressing data with NG9-1-1 requirements. Since 2015, the CAR has evolved to include active engagement with LGA's through in-depth training virtual sessions, online interactive address management mapping service, and ongoing program and technical support. This GIS data is currently integrated into the PECC, Computer-Aided Dispatch (CAD) and some other downstream systems.

The CAR is a voluntary program that is not yet fully adopted by all LGAs, leaving gaps in its utilization, especially within some rural municipalities, Indigenous communities, and northern isolated dispositions. Where these gaps exist, alternative location description methods are leveraged. The most common method is the use of the Dominion Land Survey system (DLS), legal land descriptions (LLD), which was established and used to identify rural properties prior to the introduction of the CAR. An example description of a DLS LLD would be:

Northwest 27 Township 49 Range 26 West of the Second (Meridian)

As seen above, DLS LLDs can be lengthy, which is why they are typically abbreviated for common use. In this instance, the rural property would be transcribed as:

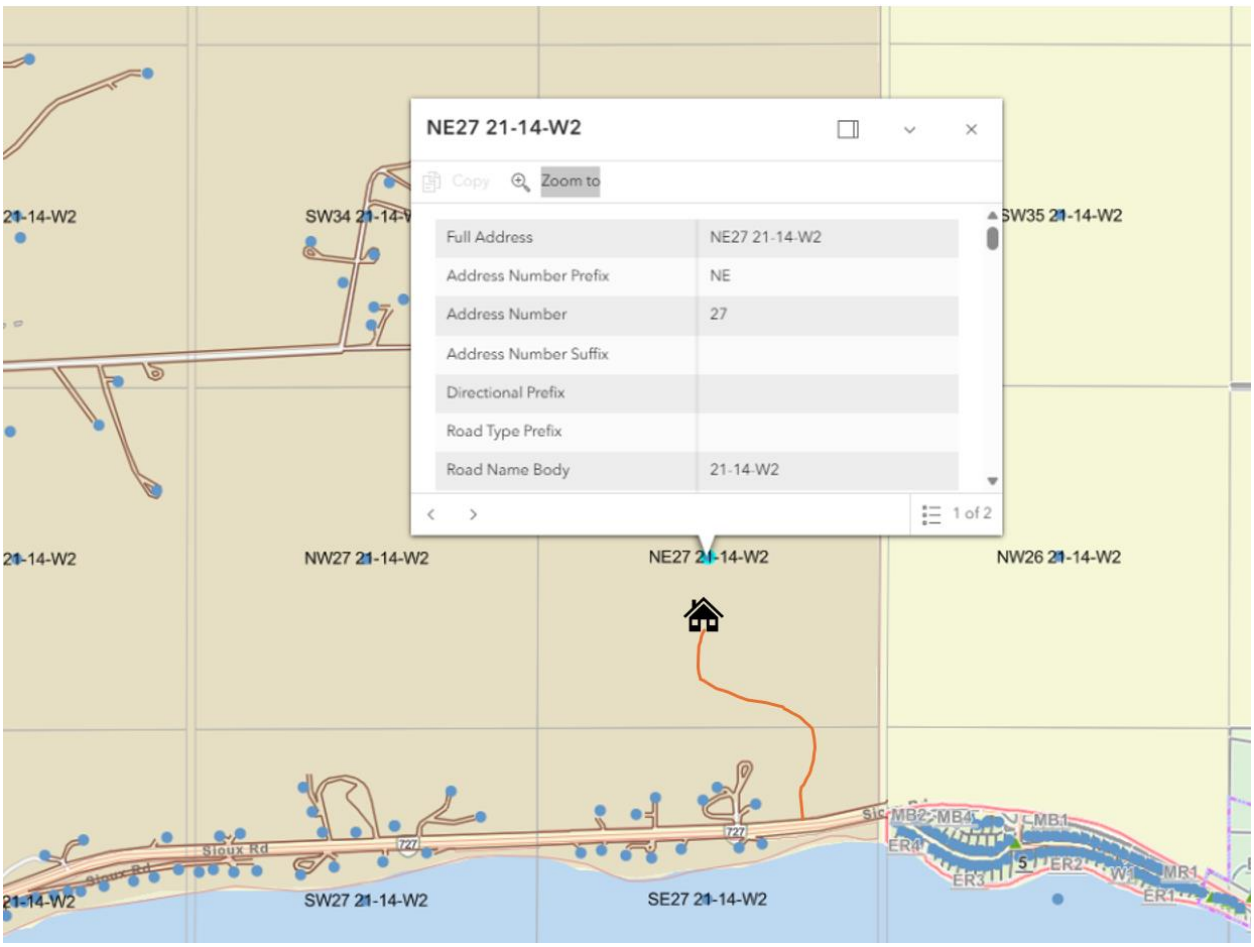
NW27-49-26-W2

In the PECC CAD system, these LLDs are spatially represented by a centroid point feature within the quarter section, which could be a considerable distance from the accessway used by emergency responders. Also, there may be multiple properties and accessways within the same quarter section.

In addition to being the NG9-1-1 network provider, SaskTel is also an ONP for Saskatchewan. Like other ONPs, they are responsible for managing customer records within their internal systems and ensuring those records can be used within 9-1-1 systems. SaskTel also uses theoretical LLD addresses placed at the centroid of each respective quarter section as placeholders when the civic address is not known or does not yet exist. This is to enable the ability to initiate a phone service, prior to the establishment of an authoritative civic address by the LGA or developer. Ideally, once an authoritative civic address has been assigned, the use of the LLD should be removed. However, until then, there will continue to be a need for a consistent and systematic approach to represent these civic locations for NG9-1-1 services.

Example:

The blue dots in the image below represent the locations for both civic addresses and the centroid of the quarter section. A new structure has been placed and is assigned the quarter section description, NE27 21-14-W2, such that the customer can be provided service, until a civic location along Sioux Rd is determined.



Option 1: Below is an example of how the address description could possibly be parsed within the SSAP data structure if existing E9-1-1 disambiguation methodologies are to be used. A downfall of the existing E9-1-1 methodology, however, arises in situations where multiple un-addressed structures exist on the same DLS LLD quarter section. In these situations, it can be difficult to disambiguate between civic locations and can lead to confusion by responders until an authoritative civic address can be assigned. Additionally, these alternative addresses do not have a have a traditional address number that would exist along a traditional road address number range. As such, address range geocoding

cannot be leveraged, and they must be disambiguated in the SSAP layer for location validation and emergency call routing.

Administrative Level (A1)	SK
Administrative Level 2 (A2)	STANDING BUFFALO INDIAN RESERVE
Additional Code	SK1212
Administrative Level (A3)	
Address Number Prefix	NE
Address Number	27
Address Number Suffix	
Address Number Complete	NE27 21-14-W2
Street Name Pre Directional	
Street Name Pre Type	
Street Name	21-14-W2
Street Name Post Type	
Street Name Post Directional	

Option 2: An alternative option, as proposed in this document, would be to leverage the location marker element to store the civic location description. Additionally, if multiple civic locations on the same DLS LLD require an address, additional location attributes can be leveraged to aid in disambiguation. The example below represents the same civic location as above, but leveraging the Location Marker field instead of forcing the DLS LLD values into traditional civic address fields:

Administrative Level (A1)	SK
Administrative Level 2 (A2)	STANDING BUFFALO INDIAN RESERVE
Additional Code	SK1212
Administrative Level (A3)	
Location Marker	NE27 21-14-W2

SPSA has a long history of managing civic addressing information for the province and transforming that data to meet the needs of 9-1-1 and emergency services. While Option #1 presented above, using E9-1-1 methodology could be made to work if required, the disambiguation methodology proposed in this document and presented as Option #2 would reduce the possibility of civic location duplication and the risk of a delayed response for emergency responders.

Newfoundland and Labrador History & Example Usage:

Newfoundland and Labrador have a province-wide Basic 9-1-1 (B9-1-1) system. This means there is no legacy E9-1-1 MSAG and similarly there was never an effort to enforce a standardized civic addressing system across the province. Existing government legislation leaves civic addressing as an optional service. The 911 Division is responsible for developing and maintaining the 9-1-1 system in the province but has no legal authority to compel local governments to establish civic addressing or to share this data with the 911 Division. The 911 Division has reached out to nearly 450 Municipalities (Muns)/Local Service Districts (LSDs) in the province to request their addressing data. At the time of writing, data has been received from 300. Of those 300 Muns/LSDs, approximately 77 sent digital GIS data (about 60 of these were via a 3rd party resource or temporary resources like a summer student or intern), with the remaining sending data via paper maps. In total, about 103 communities are deemed to be "NG-ready" with NG9-1-1-compliant civic address data (representing roughly 58% of the province's population). Work continues toward achieving NG9-1-1 readiness for all communities that have submitted civic addressing data.

There exist large areas of the province which are unincorporated (outside of a Mun or LSD). There are some unincorporated areas which have local government style services (i.e., garbage collection, snow removal, etc.) but have no local government entity to enact addressing bylaws or establish 9-1-1 service agreements. Many of these areas rely on a local association (e.g., a group of local or neighbouring landowners) to deliver those services but this style of system is not universal across all unincorporated areas.

Regardless of whether a community or area is incorporated or not, there exist many wireline/landline customer telephone number (TN) connections. These TNs have a varying level of address information and many only contain an identifier for the approximate location within the community or location where they are believed to exist. In these areas, the network provider has created pseudo-MSAGs and pseudo-addresses to tie these TNs to something resembling a civic address so that they can reconcile them with existing validation routines. Many of these TN records, however, are incomplete and/or overly generalized (e.g., a record could say "Bay Roberts – Bay Roberts" with no additional disambiguation) and are not suitable for NG9-1-1-compliant GIS data expectations. NG9-1-1-compliant GIS data expectations will result in these existing pseudo-addresses being converted into a NG9-1-1-supported location disambiguation methodology.

Example #1

A collection of homes in the unincorporated community of Brigus Junction exist as endpoint customers but have no authoritative civic address. Using the existing

authoritative information that is available for this area, the location record would appear as follows:

Administrative Level (A1)	NL
Administrative Level 2 (A2)	Placentia - St. Mary's
Additional Code	NL1234
Administrative Level (A3)	Brigus Junction
Address Number Prefix	
Address Number	
Address Number Suffix	
Address Number Complete	
Street Name Pre Directional	
Street Name Pre Type	
Street Name	Brigus
Street Name Post Type	Heights
Street Name Post Directional	
Location Marker	

This would result in duplicate “addresses” for each home in this community unless the additional location disambiguation methodology of using the Location Marker element is leveraged. Additionally, whatever description text is used in the Location Marker field, be it “House 1, House 2” or “Green House, Blue House”, etc. would have to be established jointly between the 9-1-1 authority and the originating network provider establishing the endpoint connection.

Example #2

In the following example, which has been intentionally anonymized, a collection of homes is in the A3 community of St. Paul’s, but their existing customer records have been attributed to the community of Cow Head. This is due to the issue that their local telephone exchange is associated with the community of Cow Head in the ONP’s records.

TN_NPA	TN_NXX	TL_LN_NO	Customer Surname	Customer First Name	Civic Address	Street Name	Municipality
709	123	4567	Smith	Mike	789147	COW HEAD	COW HEAD
709	123	4568	Jones	Linda	123963	COW HEAD	COW HEAD
709	123	4569	Doe	John	456258	COW HEAD	COW HEAD

Each of these records will have to be re-attributed to a community, street name, house number (if available), and /or location marker (if applicable). Again, this will require collaboration between the originating network provider, and the 9-1-1 authority.

It is expected that many of the non-traditional addressing-style civic locations will eventually phase out if and/or when these areas incorporate into a local or regional government entity which deploys a thoroughfare address-style addressing system. That process, however, may be several years in coming to fruition so methodologies will have to be used in the interim to ensure that these customers continue to receive 9-1-1 services.

9-1-1 Record Synchronisation

Various synchronisation metrics have been circulated as they relate to GIS data preparedness for NG9-1-1. "98% Match to MSAG" and other metrics often paint a picture of data quality and completeness, but they are not a guaranteed result. The goal of this synchronisation is to **ensure an equivalent, or better, call routing result during emergency calling events**. Legacy MSAG road ranges were often artificially expanded to aid in data maintenance routines. Legacy ONP customer records, and their resulting 9-1-1 records in the ALI and Selective Router databases, often contain retired, disconnected, or test records that do not need to be valid in NG9-1-1 datasets, nor should they be counted against a GIS data quality metric. The goal, during GIS data onboarding, as reinforced in the NENA GIS Data Transition Document (NENA-INF-046.1-2024) is to ensure that no legitimate customer records in legacy 9-1-1 databases should be abandoned or orphaned (e.g., going unmatched).

Additionally, the new location disambiguation options in the CLDXF-CA standard, namely the use of unique location marker or site identifiers, will not offer a direct crosswalk with existing ONP customer record database management systems. Like GIS data providers, it is unknown if ONPs will be required to change their internal systems to align with NG9-1-1 standards governing location object formatting, so long as they are able to "export" records within those systems into a NG9-1-1-compliant format for exchanges with NG9-1-1 systems. An existing system's lack of equivalent NG9-1-1 disambiguation options, however, should not force civic locations to be disambiguated by GIS data providers using only the legacy E9-1-1 "house number + street name" method.

The effort to synchronise, as ESRE0114 reinforces: "*ONPs will be required to create CLDXF-CA compliant PIDF-LOs from existing ONP customer TN and E9-1-1 MSAG records and the validation and reconciliation of those records against authoritative NG9-1-1 GIS data"*, cannot be understated. The effort between ONPs, 9-1-1 authorities, LGAs, and the GIS data providers and addressing authorities supporting them, is essential to the goal.

ONPs must be prepared to work with these key partners to ensure existing customer records are prepared for NG9-1-1, to correct/amend/remove records in error, and to be prepared to contact customers (e.g., through phone calls, billing inserts, etc.) and connect them with those key partners to ensure their locations can be disambiguated and represented in the NG9-1-1-compliant GIS data.

Conclusion & Recommendations

This contribution reinforces a civic location disambiguation methodology, initially proposed in ESCO0785, that can be used by a GIS data provider in Canada as they look to prepare their GIS data for NG9-1-1 geospatial call routing. It is flexible enough to support traditional "thoroughfare address"-style civic addressing systems, as well as non-traditional addressing systems. Additionally, it reinforces the position, initially proposed in ESCO0827, that place name data (Administrative Level polygons) should be considered a *required* GIS data layer as opposed to a *strongly recommended* layer as currently defined in the NENA NG9-1-1 GIS Data Model standard.

For the consideration of the ESWG, this contribution recommends the following:

1. That Administrative Level polygons be a required GIS layer in NG9-1-1 GIS datasets.
2. That the methodology identified in ESCO0785 become the de facto best practice for disambiguating civic locations in Canada.
    a. That NG9-1-1 network providers promote this methodology when negotiating 9-1-1 service agreements with 9-1-1 authorities and local governments
    b. That NG9-1-1 network providers support this location disambiguation methodology
3. That GIS data providers collaborate with Originating Network Providers (ONPs) to ensure 100% of appropriate ALI records can validate to an authoritative civic location record.
    a. That GIS data providers complete this effort prior to their GIS data onboarding for geospatial call routing.
    b. That ONPs complete this effort prior to a jurisdiction's GIS data onboarding for geospatial call routing.
    c. That ONPs be prepared to reach out to customers and connect them with GIS data providers and addressing authorities where required for location disambiguation.
    d. That ONPs allow for NG9-1-1-supported methodologies for disambiguating civic locations.

4. That ONPs work collaboratively with local governments and addressing authorities to reduce the need or use of unauthoritative addresses when establishing new customer connections.
5. That GIS data providers establish the GIS data management systems, workflows, and processes to ensure the timely creation, validation, and provisioning of NG9-1-1-compliant GIS data.
    a. This includes the expectations around the target 3-business day resolution of reported GIS data discrepancies.

~~~End of Document~~~
~~~